Lagrangian Variable Cosmology v0.4:

A Lagrangian Construction Reproducing the v0.3 θ-Coordinate Modulation, with Two Structural Relations Reducing the v0.3 Parameter Count

Working Paper

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Abstract

This v0.4 working paper records a Lagrangian construction whose classical solution reproduces the localised θ-coordinate modulation studied in v0.1 and v0.3. The construction is performed solely as a stress test of the framework; it is not put forward as a candidate fundamental description, and no claim is attached to the choice of action beyond the consistency of its classical solution with the previously fitted modulation. The Lagrangian is a single real scalar with the half-twist sine–Gordon potential $V(\vartheta) = m^2(1 - \cos(\vartheta/2))$. On its kink trajectory, the identification $v(\theta) \equiv T(\theta) - 1 = \varepsilon^2 \sin^2(\vartheta/2)$ yields $v(\theta) = A\,\mathrm{sech}^2((\theta - \theta_c)/w)$ with $A \equiv \varepsilon^2$ and $w \equiv 2/m$. The classical equation of motion in v admits this profile as the BPS-saturated bounce of a cubic effective potential $V_{eff}(v) = -(2/w^2)\, v^2(1 - v/A)$; the BPS condition $V_{eff}(v) = -(1/2)\, v_\theta^2$ is satisfied identically along the trajectory. Two structural relations follow without further input: (i) linearisation of the v-EOM near v = 0 fixes the width as $w = A/\ln\phi$ once $\theta_c = \ln\phi$ is imposed; (ii) the condition $\ln\phi = \phi \cdot \theta_{eq}(\Omega_m)$, where θ_{eq} is the matter–Λ equality epoch, is equivalent to $\Omega_m = 1 / (1 + e^{3 \ln\phi / \phi}) = 0.290653$. On the v0.1/v0.3 PP-excluded data set (N = 37), with locks $\theta_c = \ln\phi$, $A = (\ln\phi)^4$, $w = A/\ln\phi$, and the recursive Ω_m as above, the construction returns $\chi^2 = 56.134$ at k = 2, against $\chi^2 = 78.962$ at k = 3 for ΛCDM, giving ΔBIC = –26.44. The previously considered candidate $\Omega_m = 1/(2+\phi) = 0.27639$ returns $\chi^2 = 81.83$ ($\Delta\chi^2 = +25.7$) and is recorded as incompatible with the data at the same locks. The values $\theta_c = \ln\phi$ and the exponent k = 4 in $A = (\ln\phi)^k$ are not selected by any internal principle of the construction; $A = (\ln\phi)^4$ and $A = 1/\phi^6 = 0.0557$ differ by $\Delta\chi^2 = 0.001$ on the present data and remain indistinguishable. The construction is recorded as a consistency check; further theoretical interpretation is deferred.

1. Scope and discipline

The model class fixed since v0.1 is unchanged in this paper: a multiplicative modulation of the ΛCDM expansion rate $H(z) = H_{\Lambda CDM}(z; H_0, \Omega_m) \cdot T(z)$, with $T(z) \to 1$ at z = 0 and at high redshift, and modulation amplitude A as the only continuous parameter beyond ΛCDM in any locked candidate. The data set is the PP-excluded subset of the v0.1/v0.3 baseline (N = 37): DESI DR1 BAO (N = 11), DESI DR2 BAO (N = 13), BOSS DR12 (N = 2), eBOSS DR16 (N = 7), Planck 2018 compressed distance priors (N = 3), and SH0ES H_0 (N = 1). The Pantheon+ unbinned subset is excluded throughout; the PP-included result is recorded only where it is necessary to compare against published v0.1/v0.3 headline numbers. The chi-square pipeline is identical to the one reproduced in v0.3 and reproduces the v0.1 reference values to better than 0.005.

The goal of this work is not to establish a fundamental theory, but to determine how far the present framework can be consistently extended. The Lagrangian construction below is performed as a stress test; the resulting classical solution is compared with the previously fitted modulation; the structural relations that appear along the way are recorded; and the work is then stopped. No claim is attached to the choice of action beyond the internal consistency of its solution with v0.1/v0.3.

2. The Lagrangian and its kink solution

The action considered is that of a single real scalar field ϑ minimally coupled to gravity:

$$S = \int d^4x \sqrt{(-g)}\ [\ R/(2\kappa) + L_{matter} + L_\vartheta\ ], \quad (1)$$

$$L_\vartheta = (1/2)\ g^{\mu\nu}\ (\partial_\mu\vartheta)(\partial_\nu\vartheta) - V(\vartheta),\ V(\vartheta) = m^2\ [1 - \cos(\vartheta/2)]. \quad (2)$$

The potential $V(\vartheta)$ is periodic with period 4π, distinguishing the construction from the standard sine–Gordon potential $m^2(1 - \cos\vartheta)$ of period 2π. On the static one-soliton sector the equation of motion $\vartheta_{\theta\theta} = (m^2/2)\sin(\vartheta/2)$ admits the kink

$$\vartheta_{kink}(\theta) = 8 \arctan(\ \exp[\ (m/2)(\theta - \theta_c)\ ]\ ), \quad (3)$$

winding from $\vartheta = 0$ at $\theta \to -\infty$ through $\vartheta = 2\pi$ at $\theta = \theta_c$ to $\vartheta = 4\pi$ at $\theta \to +\infty$. Direct computation gives $\sin(\vartheta_{kink}(\theta)/2) = \mathrm{sech}[(m/2)(\theta - \theta_c)]$ and $\cos(\vartheta_{kink}(\theta)/2) = -\tanh[(m/2)(\theta - \theta_c)]$, so that

$$\sin^2(\vartheta_{kink}(\theta)/2) = \mathrm{sech}^2[(m/2)(\theta - \theta_c)]. \quad (4)$$

Identifying the modulation deviation amplitude with the squared sine of the compact-phase half-angle along the kink,

$$v(\theta) \equiv T(\theta) - 1 = \varepsilon^2 \sin^2(\vartheta_{kink}(\theta)/2), \quad (5)$$

yields $v(\theta) = A\ \mathrm{sech}^2((\theta - \theta_c)/w)$ with $A \equiv \varepsilon^2$ and $w \equiv 2/m$. Equation (4) is an algebraic consequence of (3) and is verified numerically to machine precision; no approximation is involved at this stage.

3. Equation of motion in v and the BPS structure

Working directly with $v(\theta)$, the classical equation of motion induced by the substitution (5) is autonomous in v and takes the form $v_{\theta\theta} + V'_{eff}(v) = 0$, with

$$V_{eff}(v) = -(2/w^2)\ v^2\ (1 - v/A), \quad (6)$$

a polynomial of degree three in v. The sech^2 profile is the unique trajectory satisfying $v(\theta \to \pm\infty) = 0$, $v(\theta_c) = A$, $v_\theta(\theta_c) = 0$, with the energy integral $(1/2)\ v_\theta^2 + V_{eff}(v) = 0$ along the entire trajectory; equivalently

$$V_{eff}(v(\theta)) = -(1/2)\ v_\theta(\theta)^2 \text{ (BPS-saturated bounce).} \quad (7)$$

The identities (6) and (7) hold pointwise in θ. Numerical checks at $\theta \in \{0.30, 0.40, 0.481, 0.50, 0.60, 0.70\}$ return absolute residuals below 10^{-16} for $v_{\theta\theta} + V'_{eff}(v)$ and below 10^{-17} for $V_{eff}(v) + (1/2)\ v_\theta^2$. The integral over the bounce returns $\int_{-\infty}^{+\infty} v_\theta^2\, d\theta = 16\ A^2/(15\ w)$, in agreement with the closed-form computation using $\int \mathrm{sech}^4 u\ du = 4/3$ and $\int \mathrm{sech}^6 u\ du = 16/15$.

Hubble friction in the cosmological equation of motion does not preserve (4) as an exact solution. Numerical integration of the full friction-equipped equation $v_{\theta\theta} + (3w_m/2 - 3)\ v_\theta + V'_{eff}(v) = 0$, started from $v(\theta_c) = A$, $v_\theta(\theta_c) = 0$ with $w_m(\theta) = \Omega_m\ e^{3\theta}/(\Omega_m\ e^{3\theta} + 1 - \Omega_m)$, deviates from sech^2 with a fractional error of 0.13% of A at $|\theta - \theta_c| = 0.05$ (one half-width), 1.98% at $|\theta - \theta_c| = 0.10$ (one width), and rising sharply outside the bounce window. The integrated friction over the symmetric profile vanishes by parity. The sech^2 form is therefore an exact solution of the autonomous v-EOM (Eq. 6) and an approximate solution, controlled at the percent level, of the cosmologically friction-equipped EOM within the bounce window.

4. Linearisation near v = 0 and the relation w = A/ln ϕ

Linearising $V'_{eff}(v)$ near v = 0 retains the leading term $V'_{eff}(v) \approx -(4/w^2)\, v$, so that the asymptotic decay rate of $v(\theta)$ is 2/w. Imposing the dimensionless rescaling x = v/A renders V_{eff} in the form $V_{eff}(x) = -2\,(\ln\phi)^2\, x^2\,(x - 1)$ iff w = A/ln ϕ (equivalently iff the v^2 coefficient of $V_{eff}(v)$ equals $-2(\ln\phi)^2/A^2$). The width relation

$$w = A / \ln\phi \quad (8)$$

is therefore equivalent to fixing the dimensionless coefficient of $V_{eff}(x)$ to $2(\ln\phi)^2$. Equation (8) is one constraint relating two of the three modulation parameters (θ_c, w, A); it does not by itself pin any of them.

Direct test on the data with $\theta_c = \ln\phi$ imposed and (8) imposed, leaving (H_0, Ω_m, $\Omega_b h^2$, A) free (k = 4), returns A_{fit} = 0.05469 and $w_{derived}$ = 0.11366 with χ^2 = 56.113. The reference $(\ln\phi)^k$ candidate values are $A = (\ln\phi)^4$ = 0.05362 and $w = (\ln\phi)^3$ = 0.11143; the ratios $A_{fit}/(\ln\phi)^4$ and $w_{derived}/(\ln\phi)^3$ coincide at 1.020 to four significant figures, consistent with (8). Locking in addition $A = (\ln\phi)^4$ (so that $w = (\ln\phi)^3$) returns χ^2 = 56.134 at k = 3.

5. The relation ln ϕ = $\phi \cdot \theta_{eq}(\Omega_m)$

Define the matter–Λ equality epoch in the θ = ln(1+z) coordinate by $\rho_m(\theta_{eq}) = \rho_\Lambda(\theta_{eq})$, giving $\theta_{eq}(\Omega_m) = (1/3)\ln[(1 - \Omega_m)/\Omega_m]$. Imposing the algebraic identity

$$\ln\phi = \phi \cdot \theta_{eq}(\Omega_m), \quad (9)$$

i.e. $\theta_{eq} = \ln\phi / \phi$, is equivalent to

$$\Omega_m = 1 / [\, 1 + \exp(3 \ln\phi / \phi)\,] = 0.290653\ . \quad (10)$$

Equation (9) is a self-referential identity: the modulation centre $\theta_c = \ln\phi$ and the matter–Λ equality $\theta_{eq} = \ln\phi / \phi$ stand in the ratio ϕ to one. It is recorded as an algebraic relation; no derivation of (9) from the action of Sec. 2 is offered, and none is claimed.

Direct test on the data, with $\theta_c = \ln\phi$, $A = (\ln\phi)^4$, $w = (\ln\phi)^3$, and Ω_m locked to (10), leaves only (H_0, $\Omega_b h^2$) free (k = 2) and returns χ^2 = 56.1344 against the free-Ω_m reference χ^2 = 56.1340 ($\Omega_{m,fit}$ = 0.290591). The $\Delta\chi^2$ = +0.0005 incurred by the lock is below the precision of the fit; ΔBIC = –3.61 due to the removed parameter.

For comparison, the candidate $\Omega_m = 1/(2+\phi)$ = 0.27639, which would correspond to the algebraic identity $\rho_m/\rho_\Lambda = \phi$ at $\theta = \ln\phi$, returns χ^2 = 81.83 at the same locks ($\Delta\chi^2$ = +25.7 against the free Ω_m reference, and above the ΛCDM χ^2 = 78.96). The candidate (10) and the candidate $1/(2+\phi)$ are therefore distinguished by the present data at large significance; (10) is recorded as compatible and $1/(2+\phi)$ as not.

6. Combined fit and comparison

The construction is summarised by the locks and the relation:

Quantity	Value	Status
θ_c	$\ln \phi = 0.481212$	imposed
A	$(\ln \phi)^4 = 0.053622$	imposed
w	$A / \ln \phi = (\ln \phi)^3 = 0.111432$	Eq. (8)
Ω_m	$1 / [1 + \exp(3 \ln \phi / \phi)] = 0.290653$	Eq. (10)
H_0, $\Omega_b h^2$	free	free

The free parameters of the fit are H_0 and $\Omega_b h^2$ (k = 2). On the PP-excluded data set (N = 37) the result is $\chi^2 = 56.134$, compared to the ΛCDM free-fit reference $\chi^2 = 78.962$ (k = 3). The corresponding BIC values are 63.36 and 89.79, giving ΔBIC = –26.44 in favour of the construction. The intermediate construction with $A = (\ln \phi)^4$, $w = (\ln \phi)^3$, $\theta_c = \ln \phi$ imposed but Ω_m free (k = 3) returns $\chi^2 = 56.134$, BIC = 66.97, ΔBIC = –22.83. Table 1 collects these numbers.

Model	k	χ^2	BIC	ΔBIC vs ΛCDM
ΛCDM (free H_0, Ω_m, $\Omega_b h^2$)	3	78.962	89.795	—
Sec. 2 with θ_c, A, w as $(\ln \phi)^k$; Ω_m free	3	56.134	66.967	–22.83
Same with Ω_m locked by Eq. (10)	2	56.134	63.357	–26.44
Same with $\Omega_m = 1/(2+\phi)$ instead of (10)	2	81.83	89.05	–0.74

Table 1. Goodness of fit on the PP-excluded data set (N = 37). k denotes the number of continuous parameters fitted. BIC = χ^2 + k ln N. All entries other than the ΛCDM reference share the same Lagrangian form (Eq. 2) and the same locks $\theta_c = \ln \phi$, $A = (\ln \phi)^4$, $w = (\ln \phi)^3$; they differ only in whether Ω_m is fitted, locked by Eq. (10), or locked to $1/(2+\phi)$.

7. Limits of the construction

Two of the three modulation locks of v0.1 (θ_c, w, A) are reduced to algebraic consequences of the construction: w by the linearisation argument of Sec. 4 once $\theta_c = \ln \phi$ is imposed, and Ω_m by Eq. (9) once $\theta_c = \ln \phi$ and the matter–Λ structure of the background are imposed. The remaining inputs are $\theta_c = \ln \phi$ itself and the value of A. Neither is selected by any internal principle of the action of Sec. 2; both are imposed.

The 1σ confidence interval of A on the present data, with $\theta_c = \ln \phi$, $w = A/\ln \phi$, and Ω_m locked by Eq. (10), is approximately $A \in [0.050, 0.060]$. The candidate $A = (\ln \phi)^4 = 0.0536$ and the alternative $A = 1/\phi^6 = 0.0557$ differ by $\Delta\chi^2 = 0.001$ in this fit; on the present data they are not distinguishable. The numerical near-coincidence $(\ln \phi)^3 - 2/\phi^6 \approx -2.4 \times 10^{-5}$, i.e. equality to 0.02%, propagates this ambiguity to the width: $(\ln \phi)^3$ and $2/\phi^6$ are likewise indistinguishable in the present fit.

The exponent k = 4 in $A = (\ln \phi)^k$ is selected on the present data only by the chi-square minimum across integer powers; the action of Sec. 2 does not predict this exponent. The candidates k = 3 and k = 5, with the same Lagrangian form and $\theta_c = \ln \phi$ imposed and $w = A/\ln \phi$, return χ^2 larger by $\Delta\chi^2$ = +17.6 and +6.6 respectively (these numbers are taken from the v0.4-internal scan of $(\ln \phi)^k$ exponents and are quoted here without further commentary).

The cubic effective potential $V_{eff}(v)$ of Eq. (6) is unbounded below for |v| > A and is therefore not a candidate for a globally well-posed potential. The construction of Sec. 2 is consistent in this respect because the dynamical variable is the compact phase ϑ, not v: the potential $V(\vartheta)$ of Eq. (2) is bounded and periodic, and Eq. (6) is a derived effective description on the kink trajectory only.

The relation between $V(\vartheta)$ and $V_{eff}(v)$ involves the Hubble rate at θ_c through the cosmological measure; this dependence is suppressed in the presentation above and is recorded here only as a limitation of the narrow-window approximation.

8. Stopping point

The construction has reproduced the v0.1/v0.3 modulation as the kink-sector classical solution of the action (1)–(2), reduced two of the three modulation locks of v0.1 to algebraic consequences (Eqs. 8, 10), and exhibited a fit on the PP-excluded data set with ΔBIC = –26.44 against ΛCDM at k = 2.

The values $\theta_c = \ln \phi$ and $A = (\ln \phi)^4$ remain imposed and are not selected by the construction. The data at present precision do not distinguish $A = (\ln \phi)^4$ from $A = 1/\phi^6$, and consequently do not distinguish $(\ln \phi)^3$ from $2/\phi^6$ for the width. The selection principle that would fix θ_c, A, and the exponent in $A = (\ln \phi)^k$ is not fixed within the present framework or by the present data.

Having reached this point, we intentionally stop. Further theoretical interpretation would require assumptions not supported by current data.

Appendix. Numerical reproducibility

All numerical values reported in this paper were produced with the chi-square pipeline reproduced in v0.3 from the v0.1 reference script. The Pantheon+ data files Pantheon+SH0ES.dat and Pantheon+SH0ES_STAT+SYS.cov were obtained from the official PantheonPlusSH0ES/DataRelease repository on GitHub via sparse checkout. The compressed Planck 2018 distance priors and the BAO data sets enter the PP-excluded subset (N = 37) as in v0.1 and v0.3. The Lagrangian-side identities of Sec. 2 (Eq. 4) and Sec. 3 (Eqs. 6, 7) were verified by direct symbolic substitution and by numerical evaluation at the sample points listed in the text, with absolute residuals below 10^{-16}.

Eq. (10) was tested against the data by locking Ω_m to its right-hand side and refitting H_0, $\Omega_b h^2$ by differential evolution followed by Nelder–Mead polish; the lock incurred $\Delta\chi^2 = +0.0005$ against the free-Ω_m reference. The candidate $\Omega_m = 1/(2+\phi)$ was tested in the same way and incurred $\Delta\chi^2 = +25.7$. The 1σ confidence interval on A reported in Sec. 7 was obtained by profile likelihood with Ω_m locked by Eq. (10) and the other modulation parameters tied to A by Eq. (8).